

Geometry rules matter

Signoff DRC is vast; spacing lite teaches the idea

The idea

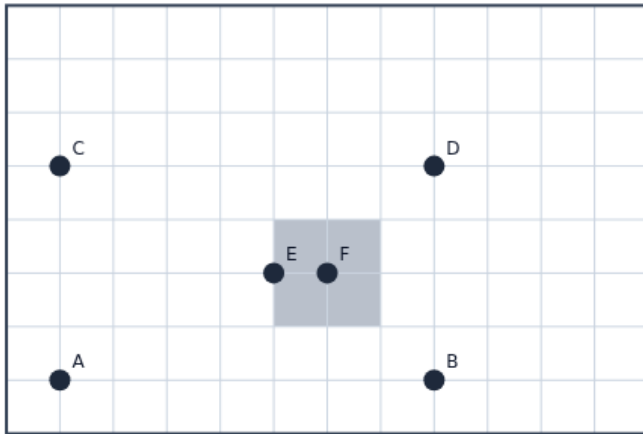
- Group segment points by layer
- For M1 pairs on the same row y
- Likewise for M2 pairs on the same column x
- Otherwise pass

Lite checker

DRC SPACING

Lite checker

Step 1 / 5 · lite · module03-01-drc-spacing



Explanation

Same-layer parallel segments one grid apart fail.

- Toy rule
- Not full deck

`minDist=1`

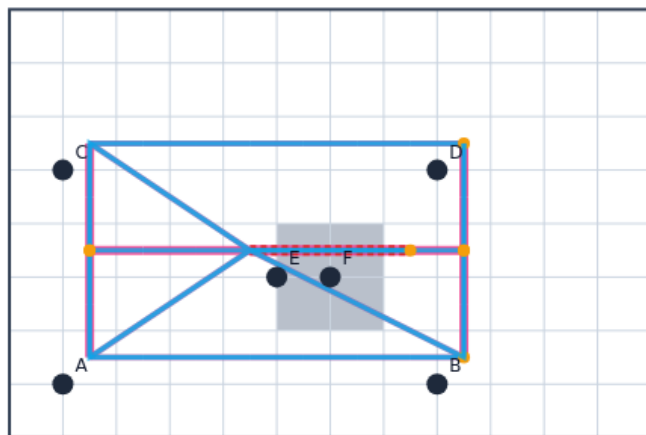
Lite checker

Spread pass

DRC SPACING

Spread pass

Step 2 / 5 · pass · module03-01-drc-spacing



Explanation

Spread layered routes pass on the toy.

- pass=true

Spread starter

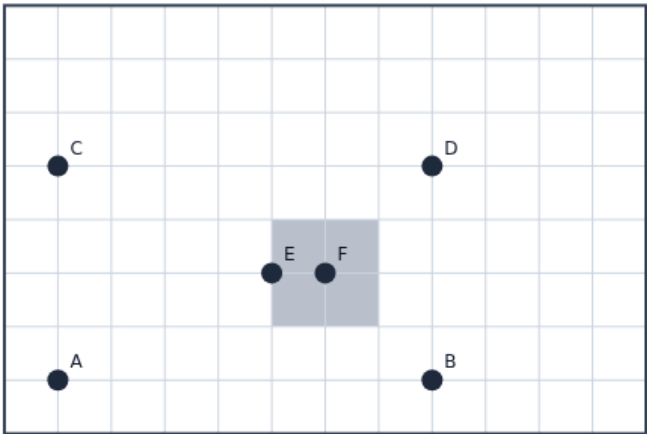
Spread pass

Violation demo

DRC SPACING

Violation demo

Step 3 / 5 · fail · module03-01-drc-spacing



Explanation

Adjacent M1 rows at y=2 and y=3 fail minDist=1.

- Fail case

Teaching

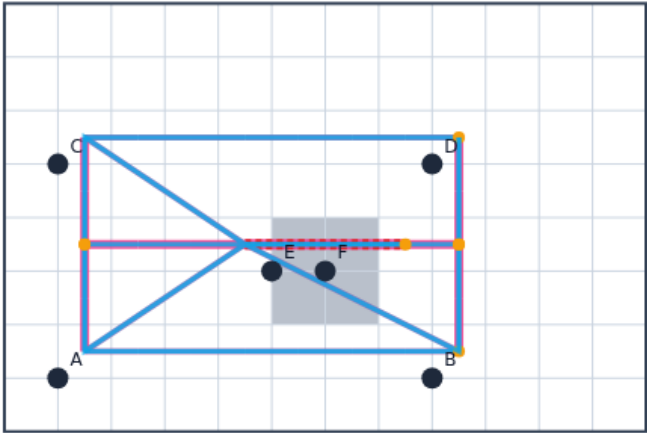
Violation demo

After route

DRC SPACING

After route

Step 4 / 5 · route · module03-01-drc-spacing



Explanation

Run checker on learner sequential routes.

- Check button

Learner routes

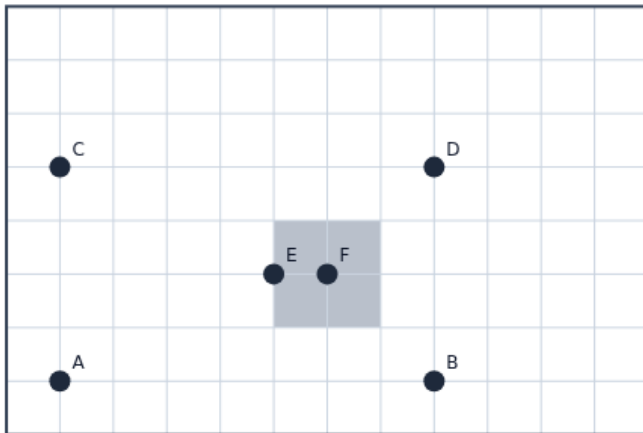
After route

Real DRC

DRC SPACING

Real DRC

Step 5 / 5 · real · module03-01-drc-spacing



Explanation

Production routers use full width/spacing/via rules.

- learn_drc next

Signoff

Real DRC

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / DRC spacing lite

INTERACTIVE TOOL

DRC spacing lite

Lite same-layer spacing check flags parallel tracks one grid apart.

Your job: run DRC on your routes. Toy checker flags adjacent parallel M1/M2 segments.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntro Spread layered passes

Pick a challenge and click **Start**. Move cells, route, then **Check**.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓ Route Lee Route layered Route A*

Rip-up detailed Check DRC

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0
DRC: pass

Browser lab starter

learn_routing — 01 drc spacing

Implement track

- Implement ``drc_spacing_lite(segments, min_dist)``
- Feed the golden failing segment set from `test_solvers` and assert pass is false

Pitfalls

- Checking cross-layer pairs, this lite rule is same-layer only
- Using Manhattan grid distance across rows for M1 horizontal segments
- Treating DRC pass as proof of tape-out readiness

Your turn

- Clear DRC spacing challenges
- Next: rip-up the hottest net and A* reroute